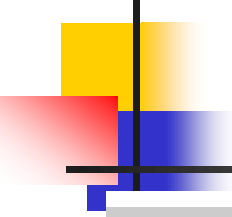


Discrete probability distributions:

Geometric, Binomial, Multinomial,
Poisson, Hypergeometric,
Negative Binomial distributions

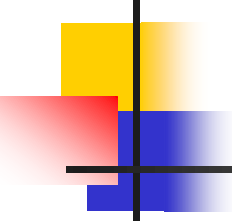


Geometric Distribution

(Geometric Distribution). A random variable X is said to have a geometric distribution with parameter p if X has pdf

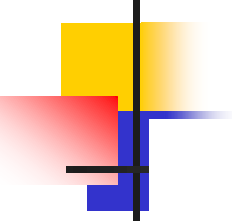
$$f(x) = \mathbb{P}(X = x) = (1 - p)^{x-1} p, \quad x = 1, 2, 3, \dots$$

We write $X \sim \text{Geom}(p)$.



Memorylessness of Geometric Dist.

- Consider the coin toss experiment. Suppose we have tossed the coin k times without a success (Heads).
- What is the probability that we need more than x additional tosses before getting a success?
- The answer is, obviously, the same as the probability that we require more than x tosses if we start from scratch, that is, $P(X > x) = (1-p)^x$ irrespective of k .
- The fact that we have already had k failures does not make the event of getting a success in the next trial(s) any more likely.
- In other words, the coin does not have a memory of what happened—hence the name memoryless property.



Mathematical Proof

Memoryless Property. Let $X \sim \text{Geom}(p)$. Then for any $x, k = 1, 2, \dots$,

$$\mathbb{P}(X > k + x \mid X > k) = \mathbb{P}(X > x) .$$

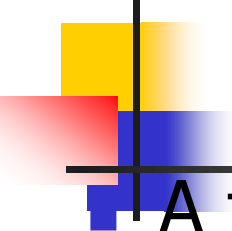
Proof. By the definition of conditional probability,

$$\mathbb{P}(X > k + x \mid X > k) = \frac{\mathbb{P}(\{X > k + x\} \cap \{X > k\})}{\mathbb{P}(X > k)} .$$

The event $\{X > k + x\}$ is a subset of $\{X > k\}$; hence their intersection is $\{X > k + x\}$. Moreover, the probabilities of the events $\{X > k + x\}$ and $\{X > k\}$ are $(1 - p)^{k+x}$ and $(1 - p)^k$, respectively. Therefore,

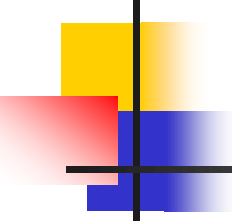
$$\mathbb{P}(X > k + x \mid X > k) = \frac{(1 - p)^{k+x}}{(1 - p)^k} = (1 - p)^x = \mathbb{P}(X > x) ,$$

as required. □



Binomial Probability Distribution

- A fixed number of observations (trials), n
 - e.g., 15 tosses of a coin; 20 patients; 1000 people surveyed
- A binary random variable
 - e.g., head or tail in each toss of a coin; defective or not defective light bulb
 - Generally called “success” and “failure”
 - Probability of success is p , probability of failure is $1 - p$
- Constant probability for each observation
 - e.g., Probability of getting a tail is the same each time we toss the coin



Binomial example

Take the example of 5 coin tosses. What's the probability that you flip exactly 3 heads in 5 coin tosses?

Binomial distribution

Solution:

One way to get exactly 3 heads: HHHTT

What's the probability of this exact arrangement?

$$\begin{aligned} &P(\text{heads}) \times P(\text{heads}) \times P(\text{heads}) \times P(\text{tails}) \times P(\text{tails}) \\ &= (1/2)^3 \times (1/2)^2 \end{aligned}$$

Another way to get exactly 3 heads: THHHT

$$\begin{aligned} &\text{Probability of this exact outcome} = (1/2)^1 \times (1/2)^3 \\ &\times (1/2)^1 = (1/2)^3 \times (1/2)^2 \end{aligned}$$

Binomial distribution

In fact, $(1/2)^3 * (1/2)^2$ is the probability of each unique outcome that has exactly 3 heads and 2 tails.

So, the overall probability of 3 heads and 2 tails is:

$(1/2)^3 * (1/2)^2 + (1/2)^3 * (1/2)^2 + (1/2)^3 * (1/2)^2 + \dots$ for as many unique arrangements as there are—but how many are there??

$$\binom{5}{3}$$

ways to
arrange 3
heads in
5 trials

$${}_5C_3 = 5!/3!2! = 10$$

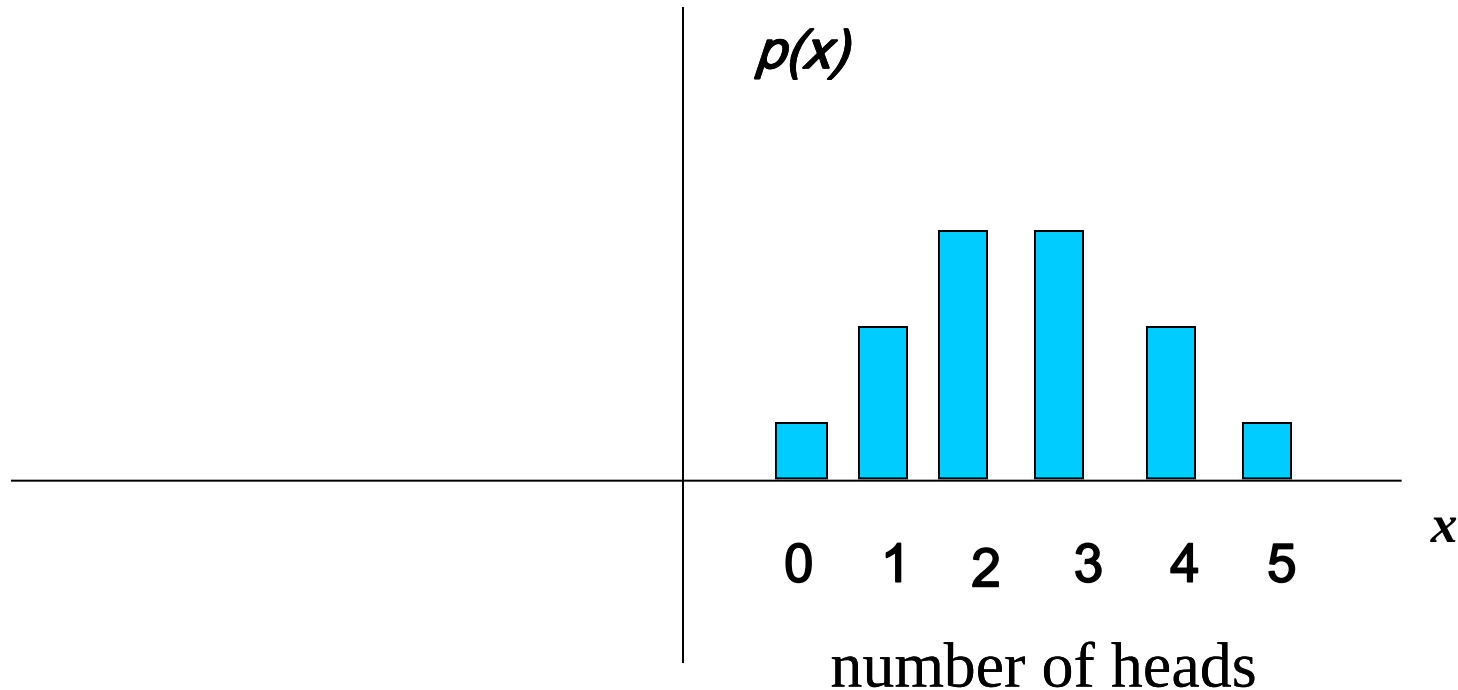
Outcome	Probability
THHHT	$(1/2)^3 \times (1/2)^2$
HHHTT	$(1/2)^3 \times (1/2)^2$
TTHHH	$(1/2)^3 \times (1/2)^2$
HTTHH	$(1/2)^3 \times (1/2)^2$
HHTTH	$(1/2)^3 \times (1/2)^2$
THTHH	$(1/2)^3 \times (1/2)^2$
HTHTH	$(1/2)^3 \times (1/2)^2$
HHTHT	$(1/2)^3 \times (1/2)^2$
THHTH	$(1/2)^3 \times (1/2)^2$
HTHHT	$(1/2)^3 \times (1/2)^2$
10 arrangements * $(1/2)^3 \times (1/2)^2$	

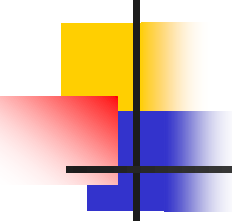
The probability
of each unique
outcome (note:
they are all
equal)

$$\therefore P(3 \text{ heads and } 2 \text{ tails}) = \binom{5}{3} * P(\text{heads})^3 * P(\text{tails})^2 =$$
$$10 \times (1/2)^5 = 31.25\%$$

Binomial distribution function:

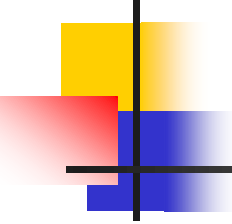
X = the number of heads tossed in 5 coin
tosses





Example 2

As voters exit the polls, you ask a representative random sample of 6 voters if they voted for proposition 100. If the true percentage of voters who vote for the proposition is 55.1%, what is the probability that, *in your sample*, exactly 2 voted for the proposition and 4 did not?



Solution:

Outcome		Probability
$\left(\begin{matrix} 6 \\ 2 \end{matrix} \right)$ ways to arrange 2 Obama votes among 6 voters	YYNNNN	$= (.551)^2 \times (.449)^4$
	NYYNNN	$(.449)^1 \times (.551)^2 \times (.449)^3 = (.551)^2 \times (.449)^4$
	NNYYNN	$(.449)^2 \times (.551)^2 \times (.449)^2 = (.551)^2 \times (.449)^4$
	NNNYYN	$(.449)^3 \times (.551)^2 \times (.449)^1 = (.551)^2 \times (.449)^4$
	NNNNYY	$(.449)^4 \times (.551)^2 = (.551)^2 \times (.449)^4$
	▪	
	▪	

15 arrangements $\times (.551)^2 \times (.449)^4$

$$\therefore P(2 \text{ yes votes exactly}) = \left(\begin{matrix} 6 \\ 2 \end{matrix} \right) \times (.551)^2 \times (.449)^4 = 18.5\%$$

Binomial distribution, generally

Note the general pattern emerging $\rightarrow$ if you have only two possible outcomes (call them 1/0 or yes/no or success/failure) in n independent trials, then the probability of exactly X “successes”=

The diagram shows the binomial probability formula $\binom{n}{X} p^X (1-p)^{n-X}$ enclosed in a purple rectangular box. Four arrows point from descriptive text to parts of the formula: one from ' n ' to ' $n = \text{number of trials}$ ', one from ' X ' to ' $X = \# \text{ successes out of } n \text{ trials}$ ', one from ' p ' to ' $p = \text{probability of success}$ ', and one from ' $1-p$ ' to ' $1-p = \text{probability of failure}$ '.

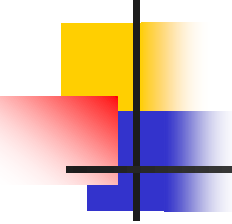
$$\binom{n}{X} p^X (1-p)^{n-X}$$

$n = \text{number of trials}$

$X = \#$
successes
out of n
trials

$p =$
probability of
success

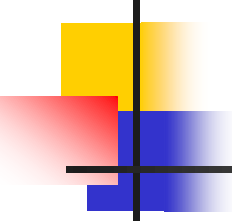
$1-p = \text{probability}$
of failure



Definitions: Binomial

- **Binomial:** Suppose that n independent experiments, or trials, are performed, where n is a fixed number, and that each experiment results in a “success” with probability p and a “failure” with probability $1-p$. The total number of successes, X , is a binomial random variable with parameters n and p .
- We write: **$X \sim \text{Bin}(n, p)$** {reads: “ X is distributed binomially with parameters n and p ”}
- And the probability that $X=r$ (i.e., that there are exactly r successes) is:

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$



Definitions: Bernoulli

Bernoulli trial: If there is only 1 trial with probability of success p and probability of failure $1-p$, this is called a Bernoulli distribution. (special case of the binomial with $n=1$)

Probability of success:

$$P(X = 1) = \binom{1}{1} p^1 (1-p)^{1-1} = p$$

Probability of failure:

$$P(X = 0) = \binom{1}{0} p^0 (1-p)^{1-0} = 1-p$$

Binomial distribution: example

- If I toss a coin 20 times, what's the probability of getting exactly 10 heads?

$$\binom{20}{10} (.5)^{10} (.5)^{10} = .176$$

Binomial distribution: example

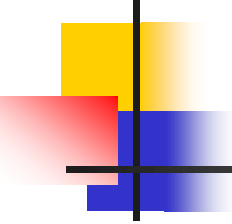
- If I toss a coin 20 times, what's the probability of getting 2 or fewer heads?

$$\binom{20}{0} (.5)^0 (.5)^{20} = \frac{20!}{20!0!} (.5)^{20} = 9.5 \times 10^{-7} +$$

$$\binom{20}{1} (.5)^1 (.5)^{19} = \frac{20!}{19!1!} (.5)^{20} = 20 \times 9.5 \times 10^{-7} = 1.9 \times 10^{-5} +$$

$$\binom{20}{2} (.5)^2 (.5)^{18} = \frac{20!}{18!2!} (.5)^{20} = 190 \times 9.5 \times 10^{-7} = 1.8 \times 10^{-4}$$

$$= 1.8 \times 10^{-4}$$



****All probability distributions are characterized by an expected value and a variance:**

If X follows a binomial distribution with parameters n and p : **$X \sim \text{Bin}(n, p)$**

Then:

$$\mu_X = E(X) = np$$

$$\sigma_X^2 = \text{Var}(X) = np(1-p)$$

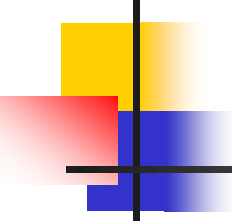
$$\sigma_X = \text{SD}(X) = \sqrt{np(1-p)}$$

Note: the variance will always lie between

$$0 * N - .25 * N$$

$p(1-p)$ reaches maximum at $p=.5$

$$P(1-p) = .25$$

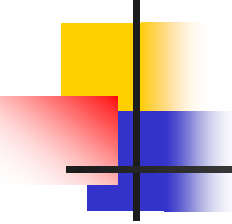


Characteristics of Bernoulli distribution

For Bernoulli ($n=1$)

$$E(X) = p$$

$$Var(X) = p(1-p)$$



Variance Proof

For $Y \sim \text{Bernoulli}(p)$

$\left\{ \begin{array}{l} Y=1 \text{ if yes} \\ Y=0 \text{ if no} \end{array} \right.$

$$\text{Var}(Y) = E(Y^2) - E(Y)^2$$

$$= [1^2 p + 0^2 (1-p)] - [1p + 0(1-p)]^2$$

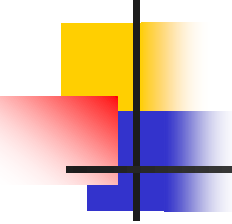
$$= p - p^2$$

$$= p(1-p)$$

For $X \sim \text{Bin}(N, p)$

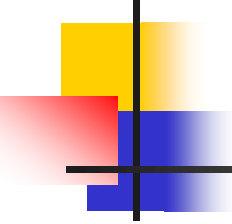
$$X = \sum_{i=1}^n Y_{\text{Bernoulli}} ; \text{Var}(Y) = p(1-p)$$

$$= \text{Var}(X) = \text{Var}\left(\sum_{i=1}^n Y\right) = \sum_{i=1}^n \text{Var}(Y) = np(1-p)$$



Recall coin toss example

- X = number of heads in 100 tosses of a coin
- $X \sim \text{Bin}(100, .5)$
- $E(X) = 100 * .5 = 50$
- $\text{Var}(X) = 100 * .5 * .5 = 25$
- $\text{SD}(X) = 5$



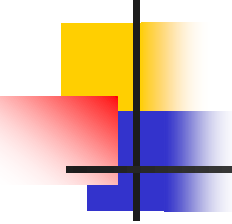
Things that follow a binomial distribution...

Cohort study (or cross-sectional):

- The number of exposed individuals in your sample that develop the disease
- The number of unexposed individuals in your sample that develop the disease

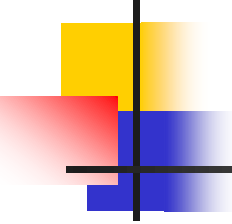
Case-control study:

- The number of cases that have had the exposure
- The number of controls that have had the exposure



Practice problems

- 1. You are performing a cohort study. If the probability of developing disease in the exposed group is .05 for the study duration, then if you sample (randomly) 500 exposed people, how many do you expect to develop the disease? Give a margin of error (+/- 1 standard deviation) for your estimate.
- 2. What's the probability that **at most** 10 exposed people develop the disease?



Answer

1. You are performing a cohort study. If the probability of developing disease in the exposed group is .05 for the study duration, then if you sample (randomly) 500 exposed people, how many do you expect to develop the disease? Give a margin of error (+/- 1 standard deviation) for your estimate.

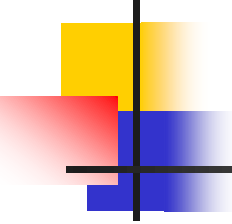
$$X \sim \text{binomial}(500, .05)$$

$$E(X) = 500 (.05) = 25$$

$$\text{Var}(X) = 500 (.05) (.95) = 23.75$$

$$\text{StdDev}(X) = \text{square root}(23.75) = 4.87$$

$$\therefore 25 \pm 4.87$$



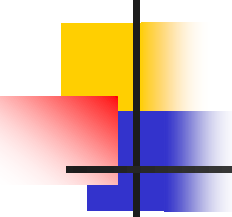
Answer

2. What's the probability that **at most** 10 exposed subjects develop the disease?

This is asking for a CUMULATIVE PROBABILITY: the probability of 0 getting the disease or 1 or 2 or 3 or 4 or up to 10.

$$P(X \leq 10) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + \dots + P(X=10) =$$

$$\binom{500}{0} (.05)^0 (.95)^{500} + \binom{500}{1} (.05)^1 (.95)^{499} + \binom{500}{2} (.05)^2 (.95)^{498} + \dots + \binom{500}{10} (.05)^{10} (.95)^{490} < .01$$



A brief distraction: Pascal's Triangle Trick

You'll rarely calculate the binomial by hand. However, it is good to know how to ...

Pascal's Triangle Trick for calculating binomial coefficients

Recall from math in your past that Pascal's Triangle is used to get the coefficients for binomial expansion...

For example, to expand: $(p + q)^5$

The powers follow a set pattern: $p^5 + p^4q^1 + p^3q^2 + p^2q^3 + p^1q^4 + q^5$

But what are the coefficients?

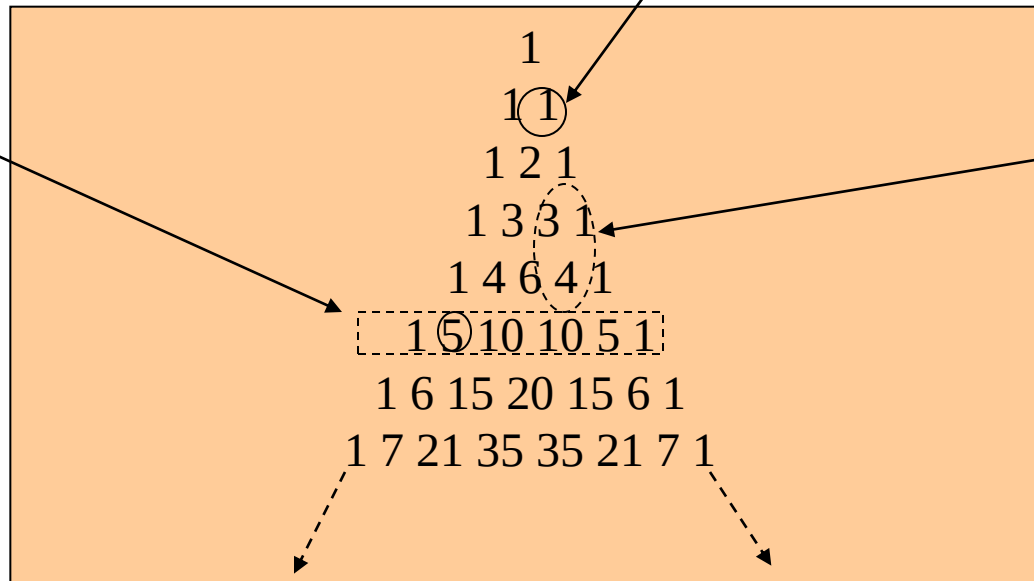
- Use Pascal's Magic Triangle...

Pascal's Triangle

To get the coefficient for expanding to the 5th power, use the row that starts with 5.

Edges are all 1's

Add the two numbers in the row above to get the number below, e.g.:
 $3+1=4$; $5+10=15$



$$(p + q)^5 = 1p^5 + 5p^4q^1 + 10p^3q^2 + 10p^2q^3 + 5p^1q^4 + 1q^5$$

Same coefficients for $X \sim \text{Bin}(5, p)$

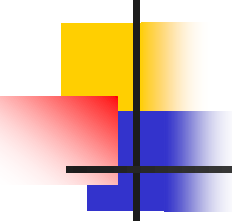
For example, $X = \#$ heads in 5 coin tosses:

X	P(X)
0	$\binom{5}{0} (.5)^0 (.5)^5$
1	$\binom{5}{1} (.5)^1 (.5)^4$
2	$\binom{5}{2} (.5)^2 (.5)^3$
3	$\binom{5}{3} (.5)^3 (.5)^2$
4	$\binom{5}{4} (.5)^4 (.5)^1$
5	$\binom{5}{5} (.5)^5 (.5)^0$

$$\begin{aligned} \binom{5}{0} &= 5!/0!5! = 1 & \binom{5}{1} &= 5!/1!4! = 5 & \binom{5}{2} &= 5!/2!3! = 5 \times 4/2 = 10 & \binom{5}{3} &= 5!/3!2! = 10 \\ \binom{5}{4} &= 5!/4!1! = 5 & \binom{5}{5} &= 5!/5!1! = 1 \end{aligned} \quad (\text{Note the symmetry!})$$

X	P(X)
0	$1(.5)^5$
1	$5(.5)^5$
2	$10(.5)^5$
3	$10(.5)^5$
4	$5(.5)^5$
5	$1(.5)^5$
$32(.5)^5 = 1.0$	

From line
5 of
Pascal's
triangle!



Relationship between binomial probability distribution and binomial expansion

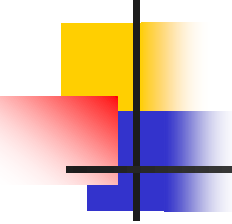
If $p + q = 1$ (which is the case if they are binomial probabilities)

then: $(p + q)^5 = (1)^5 = 1$ or, equivalently:

$$1p^5 + 5p^4q^1 + 10p^3q^2 + 10p^2q^3 + 5p^1q^4 + 1q^5 = 1$$

(the probabilities sum to 1, making it a probability distribution!)

$P(X=0)$ $P(X=1)$ $P(X=2)$ $P(X=3)$ $P(X=4)$ $P(X=5)$



Practice problems

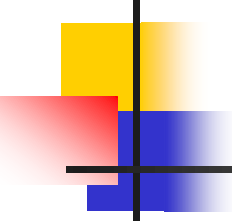
If the probability of being a smoker among a group of cases with lung cancer is .6, what's the probability that in a group of 8 cases you have less than 2 smokers? More than 5?

What are the expected value and variance of the number of smokers?

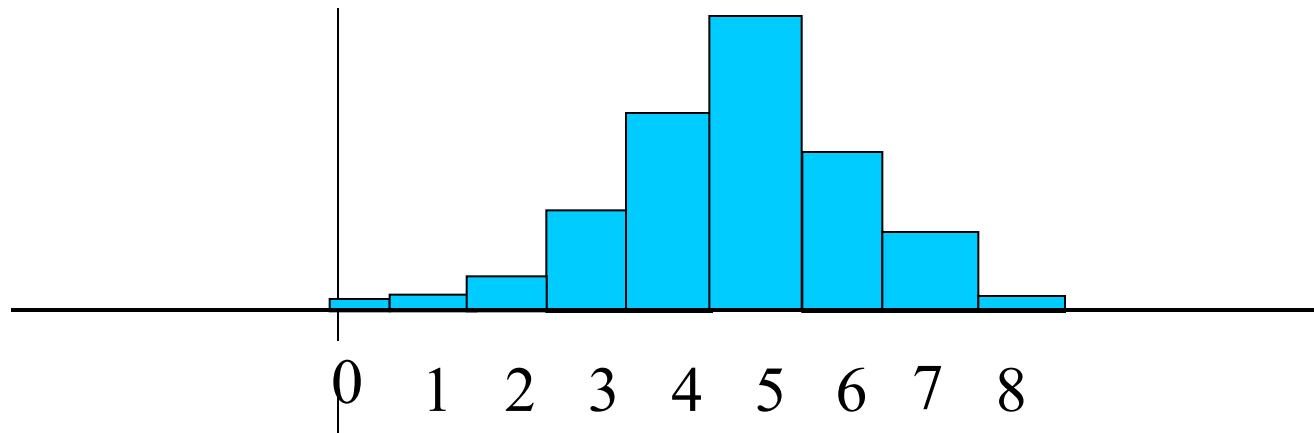
Answer

X	P(X)
0	$1(.4)^8 = .00065$
1	$8(.6)^1 (.4)^7 = .008$
2	$28(.6)^2 (.4)^6 = .04$
3	$56(.6)^3 (.4)^5 = .12$
4	$70(.6)^4 (.4)^4 = .23$
5	$56(.6)^5 (.4)^3 = .28$
6	$28(.6)^6 (.4)^2 = .21$
7	$8(.6)^7 (.4)^1 = .090$
8	$1(.6)^8 = .0168$

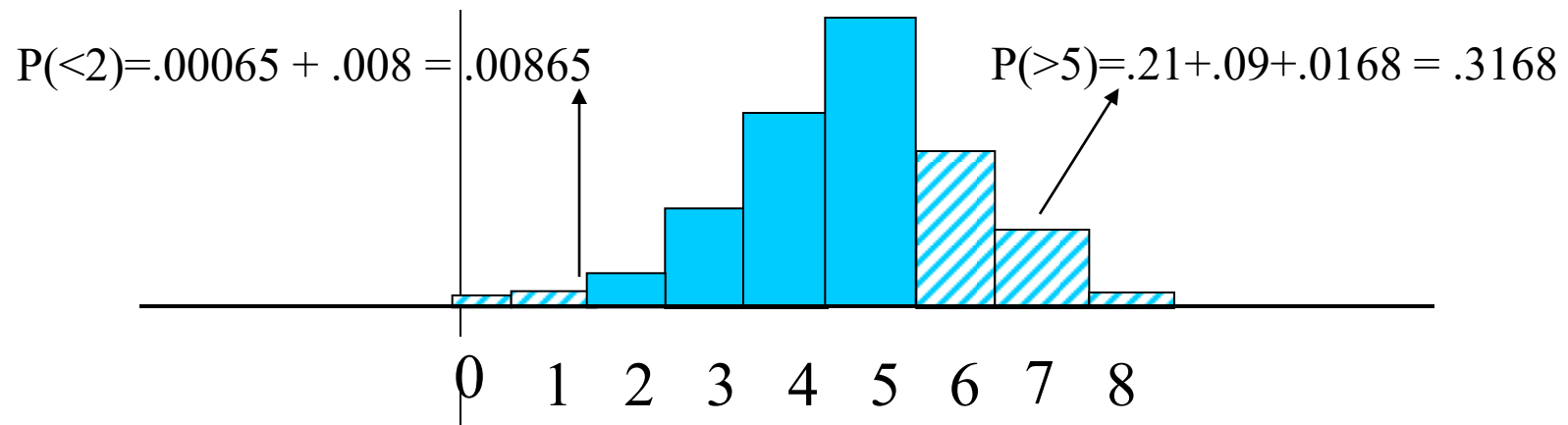
1
 1 1
 1 2 1
 1 3 3 1
 1 4 6 4 1
 1 5 10 10 5 1
 1 6 15 20 15 6 1
 1 7 21 35 35 21 7 1
 1 8 28 56 70 56 28 8 1



Answer, continued



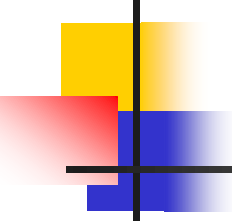
Answer, continued



$$E(X) = 8 (.6) = 4.8$$

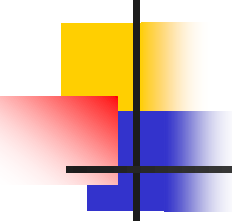
$$\text{Var}(X) = 8 (.6) (.4) = 1.92$$

$$\text{StdDev}(X) = 1.38$$



Practice problem

If Stanford tickets in the medical center 'A' lot approximately twice a week (2/5 weekdays), if you want to park in the 'A' lot twice a week for the year, are you financially better off buying a parking sticker (which costs \$726 for the year) or parking illegally (tickets are \$35 each)?



Answer

If Stanford tickets in the medical center 'A' lot approximately twice a week (2/5 weekdays), if you want to park in the 'A' lot twice a week for the year, are you financially better off buying a parking sticker (which costs \$726 for the year) or parking illegally (tickets are \$35 each)?

Use Binomial→

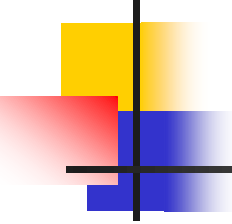
Let X be a random variable that is the number of tickets you receive in a year.

Assuming 2 weeks vacation, there are 50x2 days (twice a week for 50 weeks) you'll be parking illegally. $p=.40$ is the chance of receiving a ticket on a given day:

$X \sim \text{bin}(100, .40)$

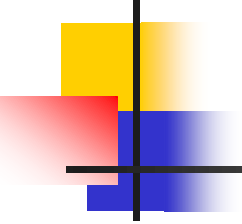
$E(X) = 100 \times .40 = 40$ tickets expected (with std dev of about 5)

$40 \times \$35 = \1400 in tickets (+/- \$200); better to buy the sticker!



Applications of Binomial Distribution

- Binomial distribution finds applications in many scientific fields.
- An industrial engineer is keenly interested in the “proportion defective” in an industrial process.
- Often, quality control measures and sampling schemes for processes are based on the binomial distribution.
- This distribution applies to any industrial situation where an outcome of a process is dichotomous and the results of the process are independent, with the probability of success being constant from trial to trial.
- The binomial distribution is also used extensively for medical and military applications.
- In both fields, a success or failure result is important. For example, “cure” or “no cure” is important in pharmaceutical work, and “hit” or “miss” is often the interpretation of the result of firing a guided missile.

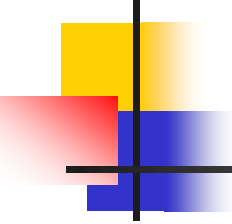


Multinomial distribution

The multinomial is a generalization of the binomial. It is used when there are more than 2 possible outcomes (for ordinal or nominal, rather than binary, random variables).

- Instead of partitioning n trials into 2 outcomes (yes with probability p / no with probability $1-p$), you are partitioning n trials into 3 or more outcomes (with probabilities: $p_1, p_2, p_3, \dots$)
 - General formula for 3 outcomes:

$$P(D = x, R = y, G = z) = \frac{n!}{x! y! z!} p_D^x p_R^y (1 - p_D - p_R)^z$$

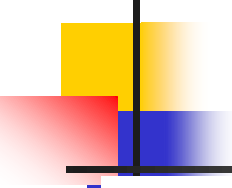


Multinomial example

Specific Example: if you are randomly choosing 8 people from an audience that contains 50% democrats, 30% republicans, and 20% green party, what's the probability of choosing exactly 4 democrats, 3 republicans, and 1 green party member?

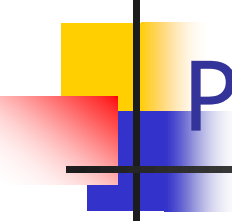
$$P(D = 4, R = 3, G = 1) = \frac{8!}{4! 3! 1!} (.5)^4 (.3)^3 (.2)^1$$

You can see that it gets hard to calculate very fast! The multinomial has many uses in genetics where a person may have 1 of many possible alleles (that occur with certain probabilities in a given population) at a gene locus.



Introduction to the **Poisson Distribution**

- Poisson distribution is for counts—if events happen at a constant rate over time, the Poisson distribution gives the probability of X number of events occurring in time T .



Poisson Mean and Variance

- Mean

$$\mu = \lambda$$

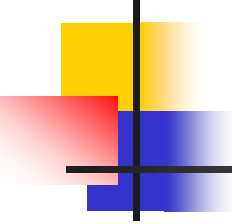
For a Poisson random variable, the variance and mean are the same!

- Variance and Standard Deviation

$$\sigma^2 = \lambda$$

$$\sigma = \sqrt{\lambda}$$

where λ = expected number of hits in a given time period



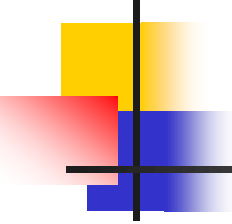
Poisson Distribution, example

The Poisson distribution models counts, such as the number of new cases of SARS that occur in women in New England next month.

The distribution tells you the probability of all possible numbers of new cases, from 0 to infinity.

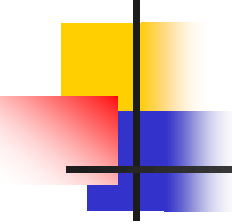
If X = # of new cases next month and $X \sim \text{Poisson}(\lambda)$, then the probability that $X=k$ (a particular count) is:

$$p(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$



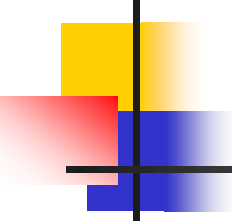
Example

- For example, if new cases of West Nile Virus in New England are occurring at a rate of about 2 per month, then these are the probabilities that: 0, 1, 2, 3, 4, 5, 6, to 1000 to 1 million to... cases will occur in New England in the next month:



Poisson Probability table

X	P(X)
0	$\frac{2^0 e^{-2}}{0!} = .135$
1	$\frac{2^1 e^{-2}}{1!} = .27$
2	$\frac{2^2 e^{-2}}{2!} = .27$
3	$\frac{2^3 e^{-2}}{3!} = .18$
4	$= .09$
5	
...	...

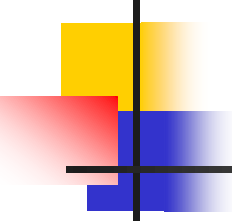


Example: Poisson distribution

Suppose that a rare disease has an incidence of 1 in 1000 person-years. Assuming that members of the population are affected independently, find the probability of k cases in a population of 10,000 (followed over 1 year) for $k=0,1,2$.

The expected value (mean) $= \lambda = .001 * 10,000 = 10$
10 new cases expected in this population per year →

$$\begin{aligned} P(X = 0) &= \frac{(10)^0 e^{-(10)}}{0!} = .0000454 \\ P(X = 1) &= \frac{(10)^1 e^{-(10)}}{1!} = .000454 \\ P(X = 2) &= \frac{(10)^2 e^{-(10)}}{2!} = .00227 \end{aligned}$$



more on Poisson...

“Poisson Process” (rates)

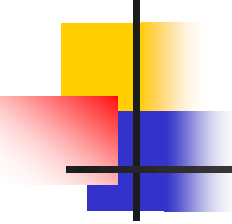
Note that the Poisson parameter λ can be given as the mean number of events that occur in a defined time period OR, equivalently, λ can be given as a rate, such as $\lambda=2/\text{month}$ (2 events per 1 month) that must be multiplied by $t=\text{time}$ (called a “Poisson Process”) $\rightarrow$

$$X \sim \text{Poisson}(\lambda t)$$

$$P(X = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

$$E(X) = \lambda t$$

$$\text{Var}(X) = \lambda t$$



Example

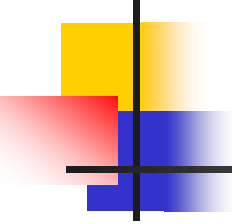
For example, if new cases of West Nile in New England are occurring at a rate of about 2 per month, then what's the probability that exactly 4 cases will occur in the next 3 months?

$X \sim \text{Poisson } (\lambda=2/\text{month})$

$$P(X = 4 \text{ in 3 months}) = \frac{(2 * 3)^4 e^{-(2*3)}}{4!} = \frac{6^4 e^{-(6)}}{4!} = 13.4\%$$

Exactly 6 cases?

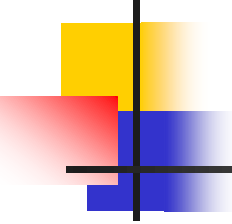
$$P(X = 6 \text{ in 3 months}) = \frac{(2 * 3)^6 e^{-(2*3)}}{6!} = \frac{6^6 e^{-(6)}}{6!} = 16\%$$



Practice problems

1a. If calls to your cell phone are a Poisson process with a constant rate $\lambda=2$ calls per hour, what's the probability that, if you forget to turn your phone off in a 1.5 hour movie, your phone rings during that time?

1b. How many phone calls do you expect to get during the movie?



Answer

1a. If calls to your cell phone are a Poisson process with a constant rate $\lambda=2$ calls per hour, what's the probability that, if you forget to turn your phone off in a 1.5 hour movie, your phone rings during that time?

$X \sim \text{Poisson} (\lambda=2 \text{ calls/hour})$

$$P(X \geq 1) = 1 - P(X=0)$$

$$P(X = 0) = \frac{(2 * 1.5)^0 e^{-2(1.5)}}{0!} \frac{(3)^0 e^{-3}}{0!} = e^{-3} = .05$$

$$\therefore P(X \geq 1) = 1 - .05 = 95\% \text{ chance}$$

1b. How many phone calls do you expect to get during the movie?

$$E(X) = \lambda t = 2(1.5) = 3$$



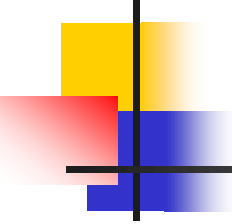
Poisson Approximation to Binomial

The Poisson distribution may be viewed as the limit of the $\text{Bin}(n, \lambda/n)$ distribution. Namely, if $X_n \sim \text{Bin}(n, \lambda/n)$, then

$$\begin{aligned}\mathbb{P}(X_n = x) &= \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \frac{\lambda^x}{x!} \frac{n \times (n-1) \times \cdots \times (n-x+1)}{n \times n \times \cdots \times n} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}.\end{aligned}$$

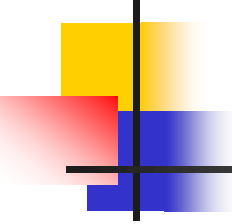
As $n \rightarrow \infty$ the second and fourth factors converge to 1 and the third factor to $e^{-\lambda}$ (this is one of the defining properties of the exponential function). Hence, we have

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = x) = \frac{\lambda^x}{x!} e^{-\lambda}.$$



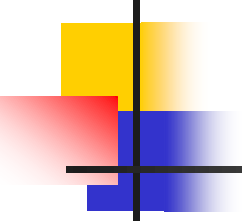
Applications of Poisson Distribution

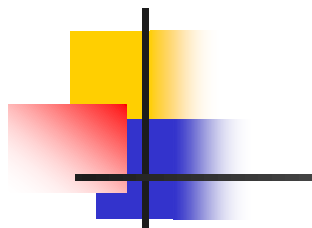
- Applications: The Poisson distribution finds applications in various fields due to its ability to model the number of events in a given time or space. Some common applications include:
- **Telecommunications:** It can be used to model the number of phone calls or text messages received by a call center or a mobile network station in a given time period.
- **Healthcare:** The Poisson distribution can be employed to analyze the number of patient arrivals at a hospital's emergency department in a specific timeframe.
- **Traffic Flow:** It can model the number of vehicles passing through a toll booth in a fixed period, helping in optimizing toll booth operations.
- **Web Server Requests:** In the context of web servers, the Poisson distribution can represent the number of requests a server receives per second.



Hypergeometric Distribution

- The simplest way to view the distinction between the binomial distribution and the hyper-geometric distribution is to note the way the sampling is done.
- The types of applications for the hyper-geometric are very similar to those for the binomial distribution.
- We are interested in computing probabilities for the number of observations that fall into a particular category.
- But in the case of the binomial distribution, independence among trials is required.
- As a result, if that distribution is applied to, say, sampling from a lot of items (deck of cards, batch of production items), the sampling must be done with replacement of each item after it is observed.

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-
- On the other hand, the hypergeometric distribution does not require independence and is based on sampling done without replacement.
 - Applications for the hypergeometric distribution are found in many areas, with heavy use in acceptance sampling, electronic testing, and quality assurance.
 - Obviously, in many of these fields, testing is done at the expense of the item being tested.
 - That is, the item is destroyed and hence cannot be replaced in the sample. Thus, sampling without replacement is necessary.

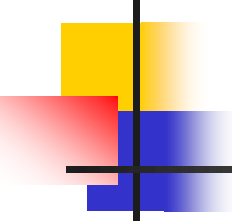


Hypergeometric
Distribution

The probability distribution of the hypergeometric random variable X , the number of successes in a random sample of size n selected from N items of which k are labeled **success** and $N - k$ labeled **failure**, is

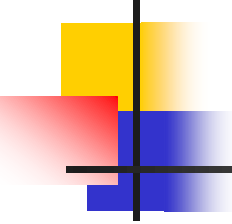
$$h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \quad \max\{0, n - (N - k)\} \leq x \leq \min\{n, k\}.$$

The range of x can be determined by the three binomial coefficients in the definition, where x and $n - x$ are no more than k and $N - k$, respectively, and both of them cannot be less than 0. Usually, when both k (the number of successes) and $N - k$ (the number of failures) are larger than the sample size n , the range of a hypergeometric random variable will be $x = 0, 1, \dots, n$.



Question

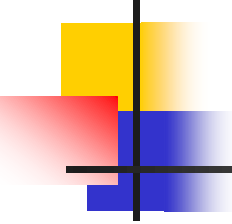
- If $N=40$, $k=39$, what will be the range of x ?
- Lots of 40 components each are deemed unacceptable if they contain 3 or more defectives. The procedure for sampling a lot is to select 5 components at random and to reject the lot if a defective is found. What is the probability that exactly 1 defective is found in the sample if there are 3 defectives in the entire lot?



Mean and Variance of Hyper Geometric Distribution

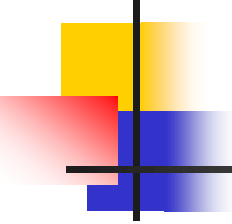
The mean and variance of the hypergeometric distribution $h(x; N, n, k)$ are

$$\mu = \frac{nk}{N} \text{ and } \sigma^2 = \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N} \left(1 - \frac{k}{N}\right).$$



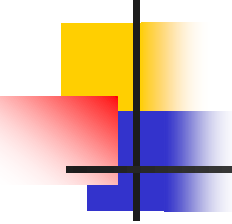
Relationship between Hypergeometric and Binomial

- There is an interesting relationship between the hypergeometric and the binomial distribution.
- As one might expect, if n is small compared to N , the nature of the N items changes very little in each draw.
- So a binomial distribution can be used to approximate the hypergeometric distribution when n is small compared to N .
- In fact, as a rule of thumb, the approximation is good when $n/N \leq 0.05$.
- Thus, the quantity k/N plays the role of the binomial parameter p . As a result, the binomial distribution may be viewed as a large-population version of the hypergeometric distribution.



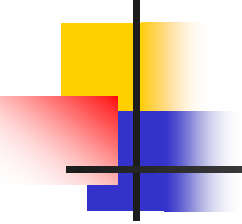
Negative Binomial Distribution

- Let us consider an experiment where the properties are the same as those listed for a binomial experiment, with the exception that the trials will be repeated until a fixed number of successes occur.
- Therefore, instead of the probability of x successes in n trials, where n is fixed, we are now interested in the probability that the k th success occurs on the x th trial.
- Experiments of this kind are called negative binomial experiments

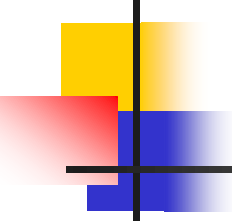


Example

- consider the use of a drug that is known to be effective in 60% of the cases where it is used.
- The drug will be considered a success if it is effective in bringing some degree of relief to the patient.
- We are interested in finding the probability that the fifth patient to experience relief is the seventh patient to receive the drug during a given week.
- Designating a success by S and a failure by F, a possible order of achieving the desired result is SFSSSFS, which occurs with probability $(0.6)(0.4)(0.6)(0.6)(0.6)(0.4)(0.6) = (0.6)^5(0.4)^2$.

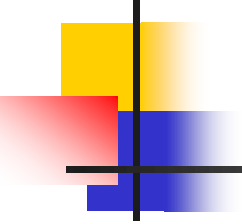
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- We could list all possible orders by rearranging the F's and S's except for the last outcome, which must be the fifth success.
 - The total number of possible orders is equal to the number of partitions of the first six trials into two groups with 2 failures assigned to the one group and 4 successes assigned to the other group
 - This can be done in 6C_4 ways. Hence, if X represents the outcome on which the fifth success occurs, then

$$P(X = 7) = \binom{6}{4} (0.6)^5 (0.4)^2 = 0.1866.$$



General Formula

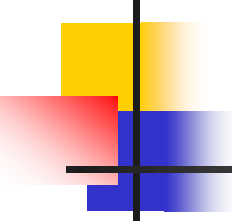
- To obtain the general formula for $b^*(x; k, p)$, consider the probability of a success on the x th trial preceded by $k - 1$ successes and $x - k$ failures in some specified order.
- Since the trials are independent, we can multiply all the probabilities corresponding to each desired outcome. Each success occurs with probability p and each failure with probability $q = 1 - p$.
- Therefore, the probability for the specified order ending in success is $p^{k-1}q^{x-k}p = p^k q^{x-k}$

- 
- The total number of sample points in the experiment ending in a success, after the occurrence of $k-1$ successes and $x-k$ failures in any order, is equal to the number of partitions of $x-1$ trials into two groups with $k-1$ successes corresponding to one group and $x-k$ failures corresponding to the other group.
 - This number is specified by the term ${}^{x-1}C_{k-1}$, each mutually exclusive and occurring with equal probability

$$p^k q^{x-k}$$

Negative Binomial Distribution If repeated independent trials can result in a success with probability p and a failure with probability $q = 1 - p$, then the probability distribution of the random variable X , the number of the trial on which the k th success occurs, is

$$b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k}, \quad x = k, k+1, k+2, \dots$$



Negative Binomial and Geometric

- When $k=1$, negative binomial dist. Reduces to geometric distribution.

Definition

The **probability generating function (PGF)** of a discrete random variable is given by:

$$G_X(t) = \mathbb{E}(t^X) = \sum_x t^x \mathbb{P}(X = x)$$

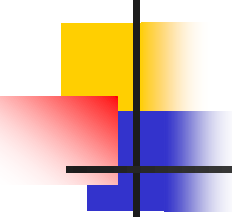
where t is known as a dummy variable.

This comes from the formula of the expectation of a function of a discrete random variable:

$$\mathbb{E}(g(X)) = \sum_x g(x) \mathbb{P}(X = x)$$

where $g(X) = t^X$.

From the formula, you can see that each term of the PGF is a t^x term with a coefficient. The value of the exponent, x , corresponds to a value that the random value can take and the coefficient of each t^x term corresponds to the probability of the random variable taking the value of the exponent.



Properties of Geometric distribution

Example 1.1 (Properties of the Geometric Distribution). Let $X \sim \text{Geom}(p)$. Then,

1. $\mathbb{E}X = 1/p$,
2. $\text{Var}(X) = (1-p)/p^2$,
3. the PGF is

$$G(z) = \frac{z p}{1 - z(1 - p)}, \quad |z| < \frac{1}{1 - p}. \quad (2.13)$$

Proof. The PGF of X follows from

$$G(z) = \sum_{x=1}^{\infty} z^x p(1-p)^{x-1} = z p \sum_{k=0}^{\infty} (z(1-p))^k = \frac{z p}{1 - z(1-p)},$$

using the well-known result for *geometric sums*: $1 + a + a^2 + \cdots = (1-a)^{-1}$, for $|a| < 1$.